CS & IT ENGINEERING

Digital Logic

Optimisation

Lecture No. 05

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Recap of Previous Lecture

Topic

Min-terms

Max-terms

K-Maps

Topics to be Covered

Topic

K-maps

Practice Questions

K-map

↳ Gray Code → unity HD

↳ Cyclic

↳ Reflecting

Gray Code

A	B
0	0
0	1
1	1
1	0

Deci	A	B	C
0	0	0	0
1	0	0	1
2	0	1	1
3	0	1	0
4	1	1	0
5	1	1	1
6	1	0	1
7	1	0	0

$F(A,B)$

A \ B	0	1
0	00_0	01_1
1	10_2	11_3

$F(A,B,C)$

A \ BC	00	01	11	10
0	000_0	001_1	011_3	010_2
1	100_4	101_5	111_7	110_6

0	1	3	2
4	5	7	6

SOP

$f(A,B,C,D)$

AB \ CD	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	CD 11	$C\bar{D}$ 10
$\bar{A}\bar{B}$ 00	0	1	3	2
$\bar{A}B$ 01	4	5	7	6
AB 11	12	13	15	14
$A\bar{B}$ 10	8	9	11	10

$\bar{A}\bar{B}CD$

$AB\bar{C}D$

(a, b)

Simplification

└→ (Grouping)

Imp: Groups can only be in order of 2^n.

Group size

$1 \rightarrow 2^0 \rightarrow$ Reducing 0 variables.

(pair) $2 \rightarrow 2^1 \rightarrow$ " 1 variable

(quad) $4 \rightarrow 2^2 \rightarrow$ " 2 variables

(octa) $8 \rightarrow 2^3 \rightarrow$ " 3 variables

$16 \rightarrow 2^4 \rightarrow$ 4 variables

$2^0 \rightarrow 1$

$2^1 \rightarrow 2$

$2^2 \rightarrow 4$

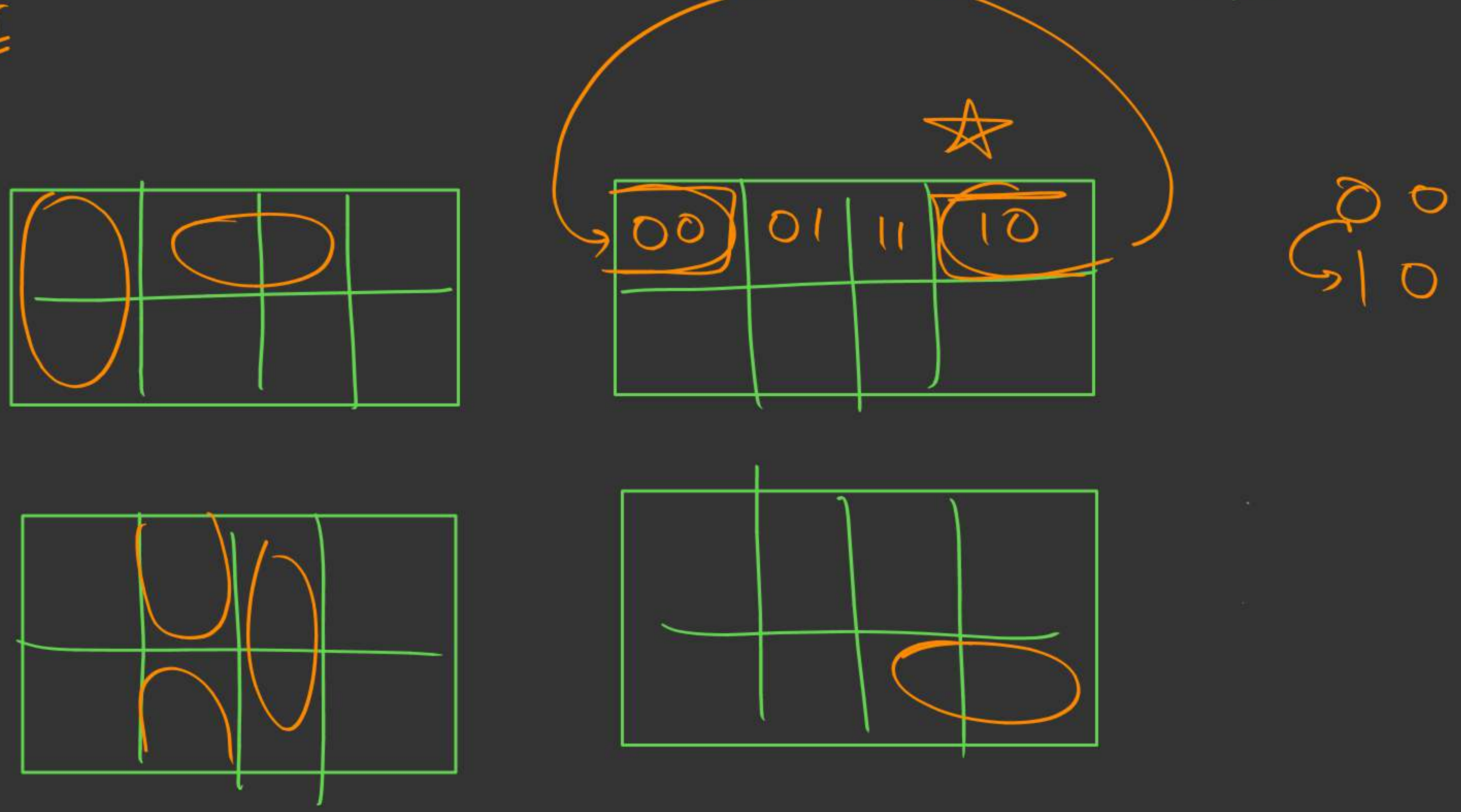
$2^3 \rightarrow 8$

$\vdots \rightarrow 16$

Grouping $\longrightarrow$ (neighbours shall have 1 bit diff)

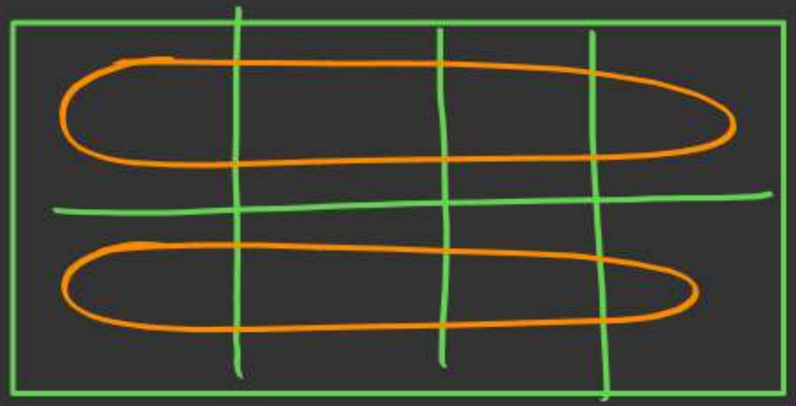
Grouping

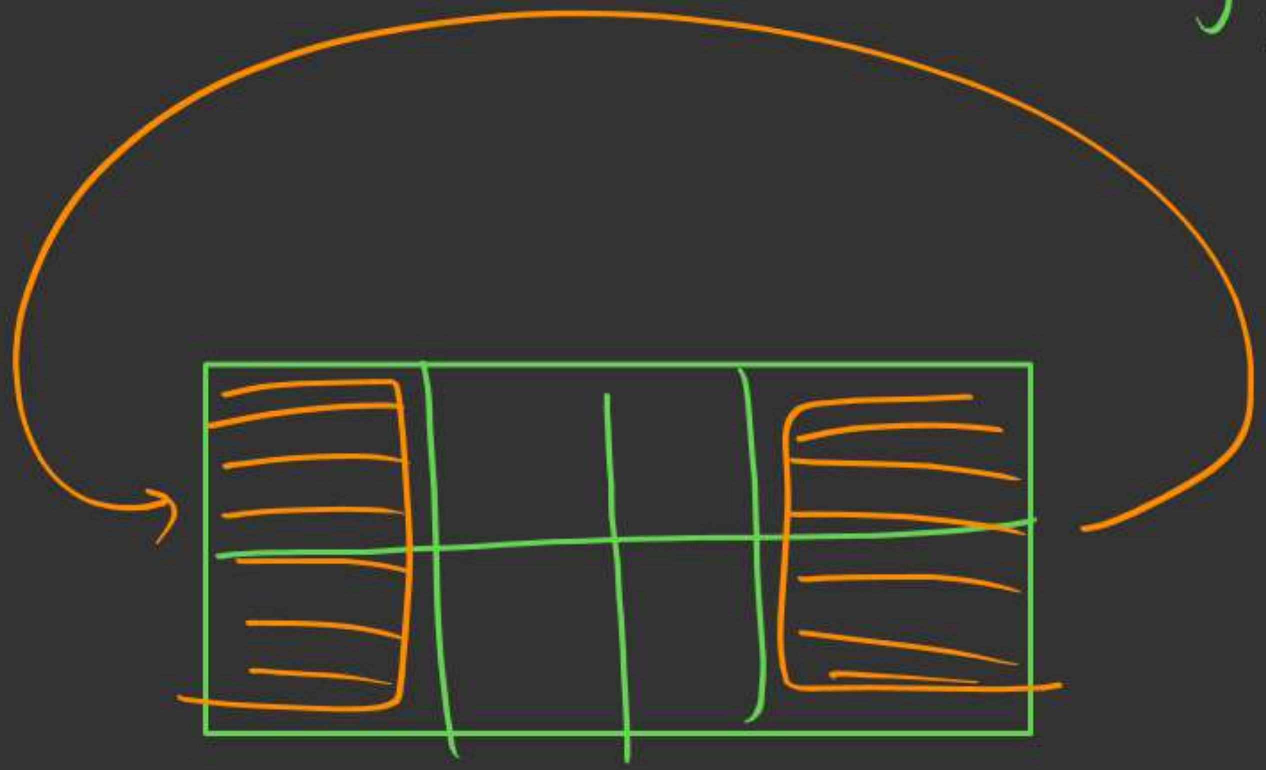
pair

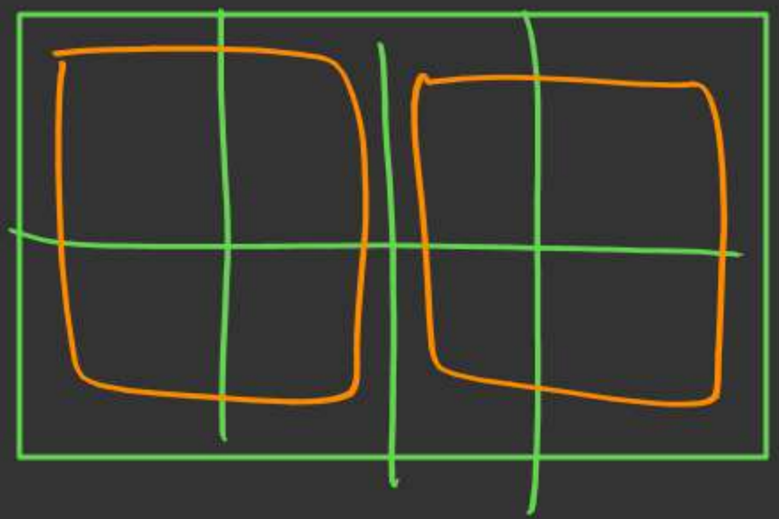


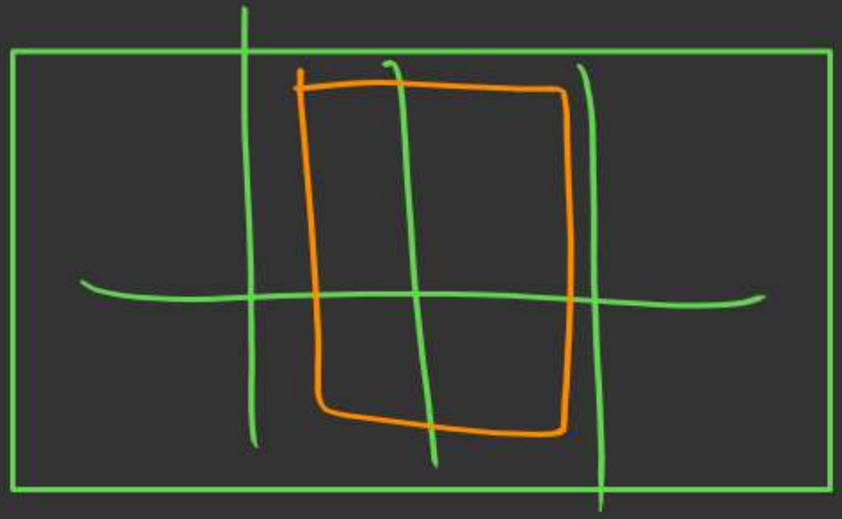
Quad

Grouping

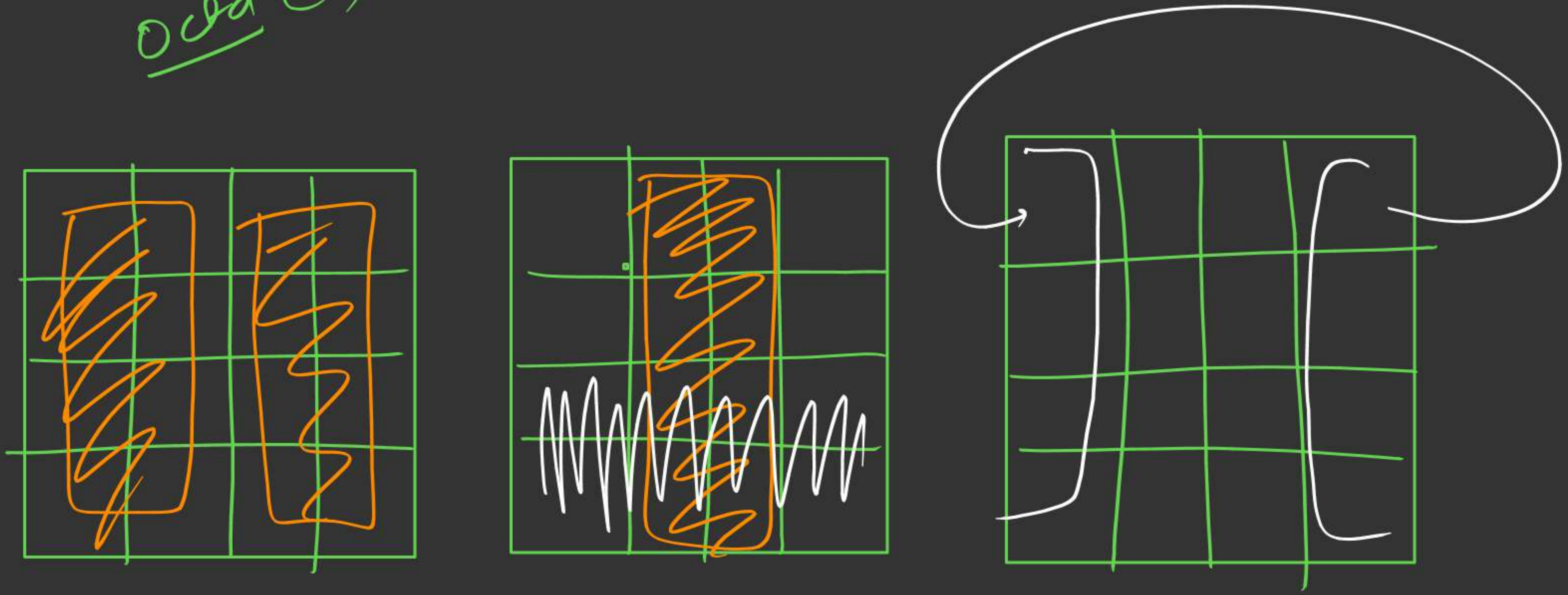


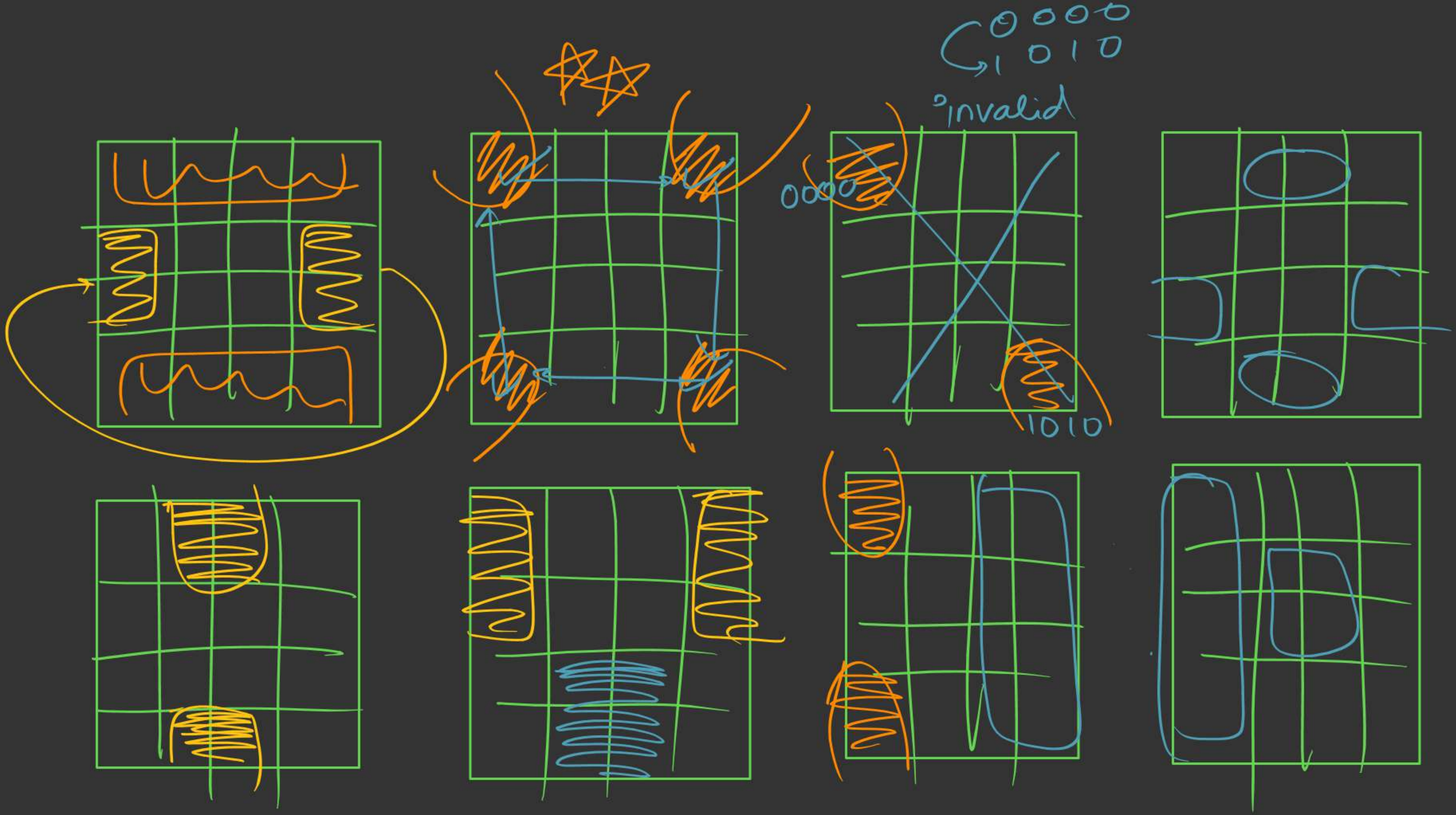






Octa (8)




0000
1010
invalid
0000
1010

Grouping of 1's
(SOP)

1 turn

K-map ⟶ minimize the expression

count of terms will minimize

Logic : Prefer Lesser no. of Groups.

and Prefer as Large group as possible.

minimizes the no. of variables within a term

$\Sigma m \rightarrow SOP$

eg:- $f(P,Q) = PQ + \bar{P}\bar{Q} + \bar{P}Q$

$= \Sigma m(0,1,3)$

		$\bar{Q}$ 0	Q 1
$\bar{P}$	0	1_{0}	1_{1}
P	1	$_{2}$	1_{3}

$\bar{P} \cdot 1$

$Q \cdot 1$

$$\boxed{f(P,Q) = \bar{P} + Q}$$

0 1 0

$$f(p,q) = \bar{p}\bar{q} + \underbrace{pq + \bar{p}q}$$

$$= \bar{p}\bar{q} + q(p+\bar{p})$$

$$= \bar{p}\bar{q} + q$$

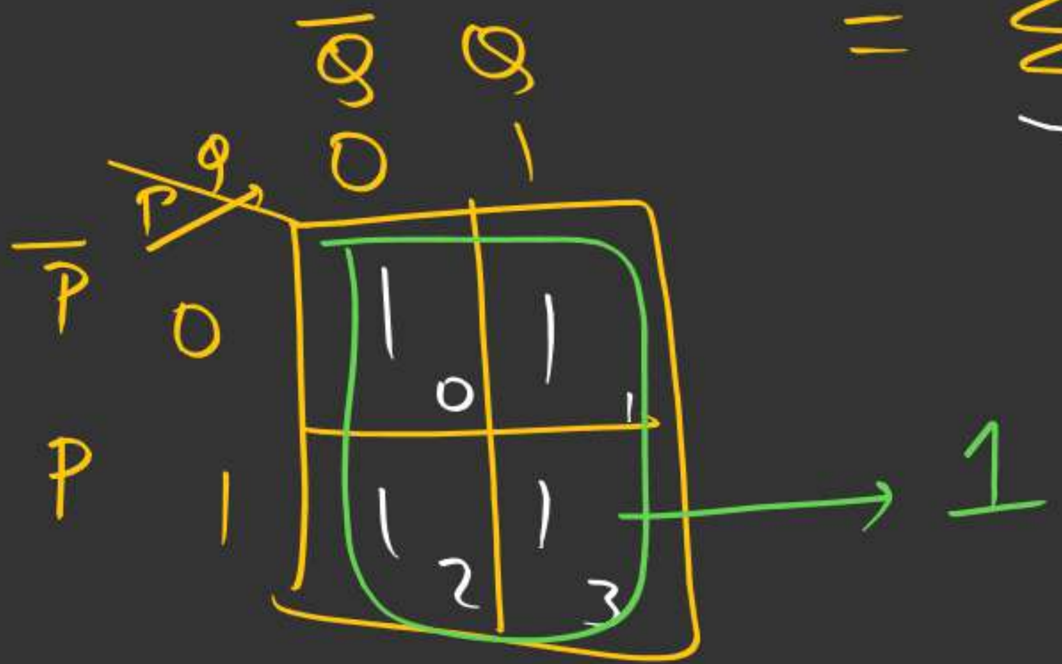
$$= (\bar{p}+q)(q+\bar{q})$$

$$= \underline{\underline{\bar{p}+q}}$$

(2)

$$f(P,Q) = PQ + \bar{P}Q + \bar{P}\bar{Q} + P\bar{Q}$$

$$= \Sigma m(0,1,2,3)$$



③

$$F(P,Q,R) = \sum m(0,1,3,5,6,7)$$

$P \backslash QR$		$\bar{Q}\bar{R}$ 00	$\bar{Q}R$ 01	QR 11	$Q\bar{R}$ 10
$\bar{P}$	0	1 $_0$	1 $_1$	1 $_3$	$_2$
P	1	$_4$	1 $_5$	1 $_7$	1 $_6$

$\bar{P}\bar{Q}$

R

PQ

$$F(P,Q,R) = \bar{P}\bar{Q} + R + PQ$$

SCF SOP

2°

④ $f(P,Q,R) = \sum m(0,1,3,6,7)$

①

$P \backslash QR$	$\bar{Q}\bar{R}$ 00	$\bar{Q}R$ 01	QR 11	$Q\bar{R}$ 10
$\bar{P}$ 0	1	1	1	
P 1			1	1

$\bar{P}\bar{Q}$ $\quad \bar{P}R \quad PQ$

$F = (\bar{P}\bar{Q} + \bar{P}R + PQ)$

②

$P \backslash QR$	00	01	11 (QR)	10
$\bar{P}$ 0	1	1	1	
P 1			1	1

$\bar{P}\bar{Q} \quad QR \quad PQ$

$F = \bar{P}\bar{Q} + QR + PQ$

③

$P \backslash QR$	$\bar{Q}\bar{R}$ 00	$\bar{Q}R$ 01	QR 11	$Q\bar{R}$ 10
$\bar{P}$ 0	1	1	1	
P 1			1	1

$\bar{P}\bar{Q} \quad \bar{P}QR \quad PQ$

$F = (\bar{P}\bar{Q} + \bar{P}QR + PQ)$

semi minimized expression

⑤ $F(P,Q,R) = \sum m(0,2,4,6)$

$P \backslash QR$	$\bar{Q}\bar{R}$	$\bar{Q}R$	QR	$Q\bar{R}$
$\bar{P}$	1			1
P	1			1

$F = \bar{R}$

$= \checkmark$

⑥ $F(P,Q,R,S) = \sum m(0,2,4,6,10,11,13,15)$

$PQ \backslash RS$		$\bar{R}\bar{S}$ 00	$\bar{R}S$ 01	RS 11	$R\bar{S}$ 10
$\bar{P}\bar{Q}$	00	1 (0)	(1)	(3)	1 (2)
$\bar{P}Q$	01	1 (4)	(5)	(7)	1 (6)
PQ	11	(12)	1 (13)	1 (15)	(14)
$P\bar{Q}$	10	(8)	(9)	1 (11)	1 (10)

$\bar{P}\bar{S}$

PQS

$P\bar{Q}R$

$(\bar{P}\bar{S} + PQS + P\bar{Q}R)$

(7) $f(P,Q,R) = \Sigma m(0,3,5,6)$

	$\bar{Q}\bar{R}$	$\bar{Q}R$	QR	$Q\bar{R}$
$\bar{P}$	1		1	
P		1		1

$\bar{P}\bar{Q}\bar{R}$

$\bar{P}QR$

$P\bar{Q}R$

$PQ\bar{R}$

$$F = (\bar{P}\bar{Q}\bar{R} + \bar{P}QR + PQ\bar{R} + P\bar{Q}R)$$

(8) $F(P,Q,R,S) = \sum m(0,1,2,4,6,9,10,11,12,13,15)$

PQ \ RS	$\bar{R}\bar{S}$ 00	$\bar{R}S$ 01	RS 11	$R\bar{S}$ 10
$\bar{P}\bar{Q}$ 00	1 (0)	1 (1)	(3)	1 (2)
$\bar{P}Q$ 01	1 (4)	(5)	(7)	1 (6)
PQ 11	1 (12)	1 (13)	1 (15)	(14)
$P\bar{Q}$ 10	(8)	1 (9)	1 (11)	1 (10)

$\bar{P}\bar{Q}\bar{R}$

$\bar{P}\bar{S}$

$PQ\bar{R}$

PS

$P\bar{Q}R$

$$F = \bar{P}\bar{S} + P\bar{Q}R + PS + PQ\bar{R} + \bar{P}\bar{Q}\bar{R}$$

(Q) H.W

$$f(P,Q,R,S) = \sum m(1,5,6,7,11,12,13,15)$$

2 mins Summary

K-map

Questions

THANK - YOU